Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 25 shows an object at the bottom of a beaker of water.

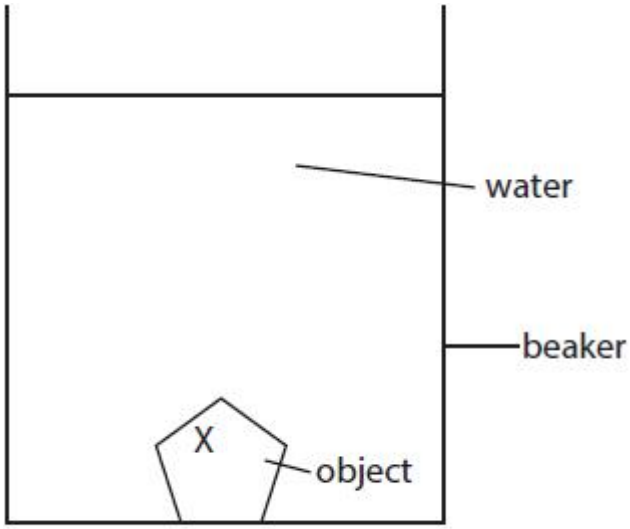
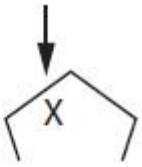


Figure 25

Which diagram shows the direction of the force exerted by the water on the object at point X?

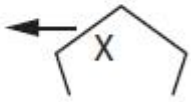
(1)

☐ **A**

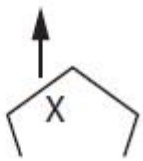


☐ **B**

☐ **C**



☐ **D**



(Total for question = 1 mark)

Q2.

The student measures the extension of the spring as he adds different loads (loading).

He then measures the extension of the spring as he takes the loads off (unloading).

He then repeats the investigation using a rubber band instead of the spring.

The tables in Figure 4 show his results.

spring		
load in N	extension in mm	extension in mm
	loading	unloading
0	0	0
1	20	20
2	40	40
3	60	60

rubber band		
load in N	extension in mm	extension in mm
	loading	unloading
0	0	0
1	14	25
2	33	42
3	60	60

Figure 4

State **two** similarities and **two** differences between the results for the spring and the results for the rubber band.

(4)

Similarity 1

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.....

Similarity 2

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.....

Difference 1

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.....

Difference 2

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.....

(Total for question = 4 marks)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 3 shows a diver swimming in a lake.

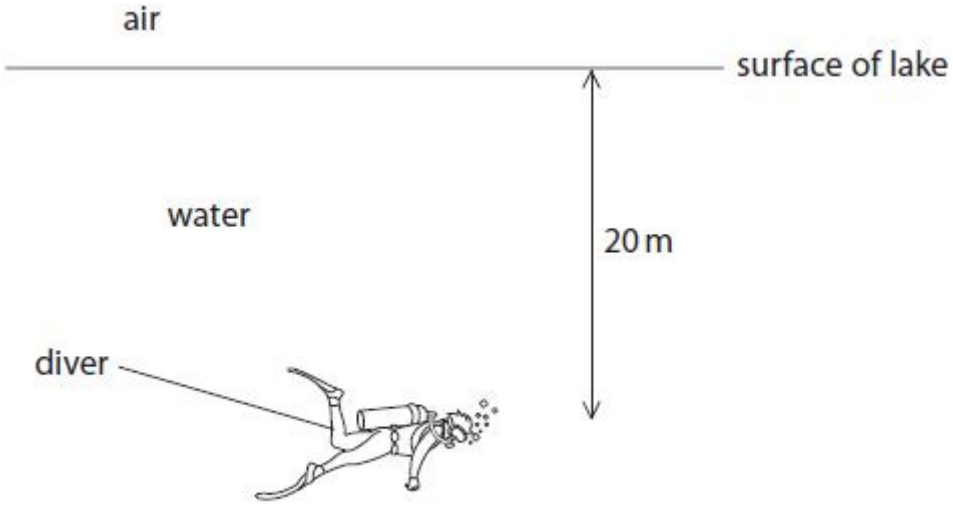


Figure 3

The pressure on the diver is due to both the water above him and the Earth's atmosphere.

The pressure of air on the surface of the water is one atmosphere. 10 m of water is equivalent to one atmosphere.

How many atmospheres of pressure will be on the diver at a depth 20 m?

(1)

☐ **A** 1
☐ **B** 2
☐ **C** 3
☐ **D** 4

(Total for question = 1 mark)

Q4.

Figure 25 is a graph showing how the atmospheric pressure changes with the height above sea level on the Earth's surface.

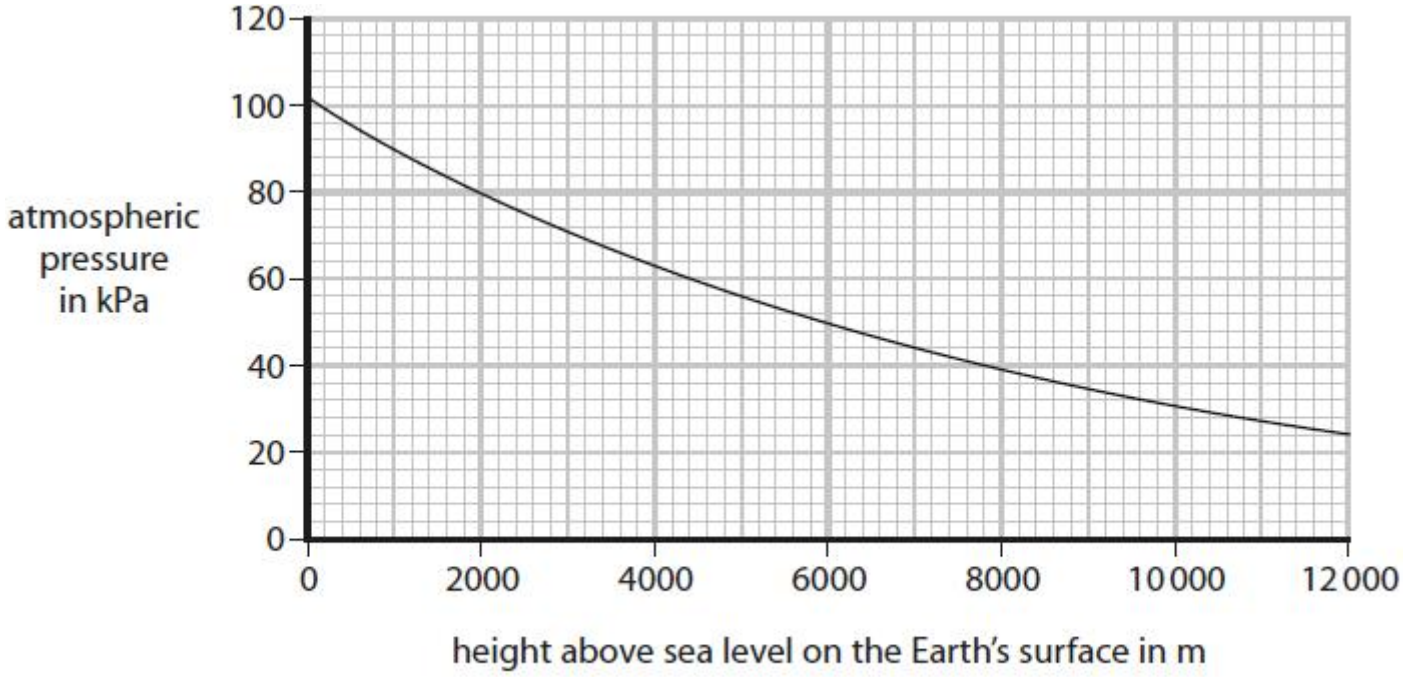


Figure 25

(i) An aeroplane descends from 6000 m to 2000 m.

Use the graph to find the change in atmospheric pressure as the aeroplane descends.

(2)

change in pressure = ... kPa

(ii) Suggest **one** reason why the atmospheric pressure is greater at 2000 m than at 6000 m.

(1)

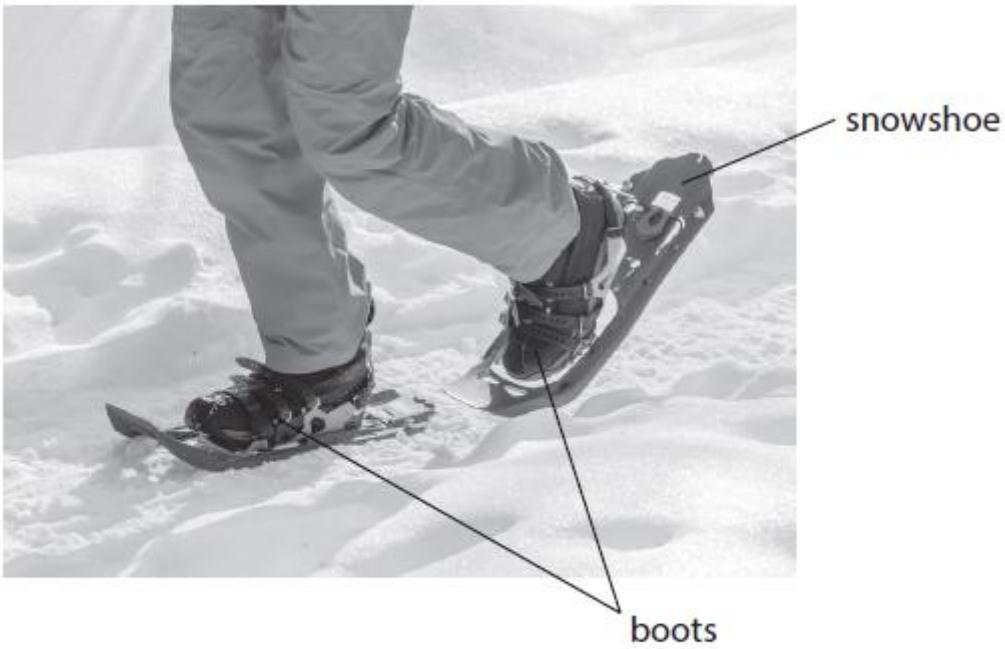
.....

.....

(Total for question = 3 marks)

Q5.

Figure 7 shows a person wearing snowshoes.



(Source: Pearson Asset Library)

Figure 7

Snow shoes are designed for walking on deep, soft snow.

Explain why they are better suited for this than boots.

(2)

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.....

.....

(Total for question = 2 marks)

Q6.

A balloon is filled with helium when it is on the ground.

The balloon is released and it rises through the atmosphere.

Explain what happens to the size of the balloon as it rises through the atmosphere.

(3)

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(Total for question = 3 marks)

Q7.

A scuba diver is on a sandy beach.

She checks her compressed air cylinders before a dive.
She has two identical steel cylinders, A and B.
Each cylinder contains the same amount of compressed air.

Figure 25 shows the diver's cylinders.

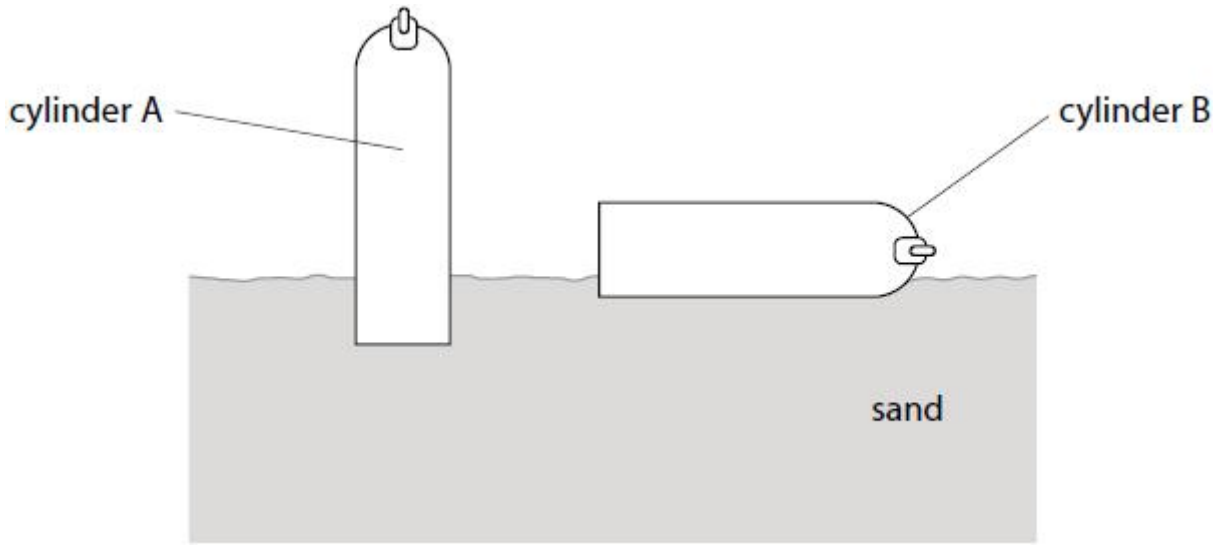

Figure 25

Explain why cylinder A sinks further into the sand than cylinder B.

Use ideas about pressure, force and area in your answer.

(4)

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(Total for question = 4 marks)

Q8.

Figure 28 is a model representing molecules of the Earth's atmosphere.

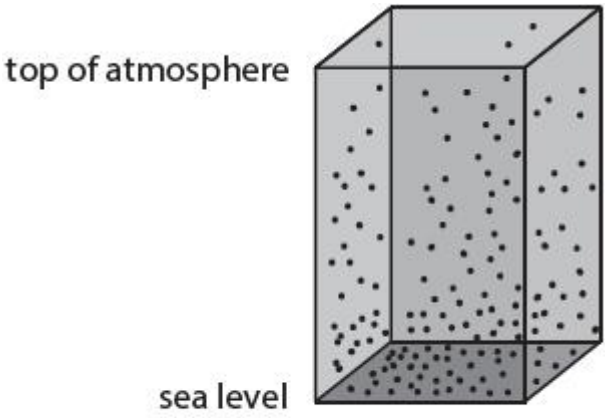

Figure 28

Use Figure 28 to explain how the density of the air varies with height above sea level.

(2)

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.....

(Total for question = 2 marks)

Q9.

Figure 26 shows two drawings of the same person on a bed.

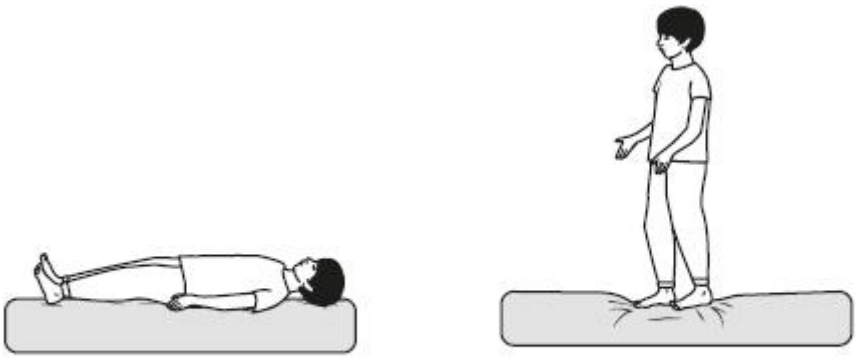

Figure 26

Explain why the person exerts a different pressure on the bed when standing up than when lying down.

(2)

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.....

.....

.....

(Total for question = 2 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 10 shows a toy used to launch a ball.

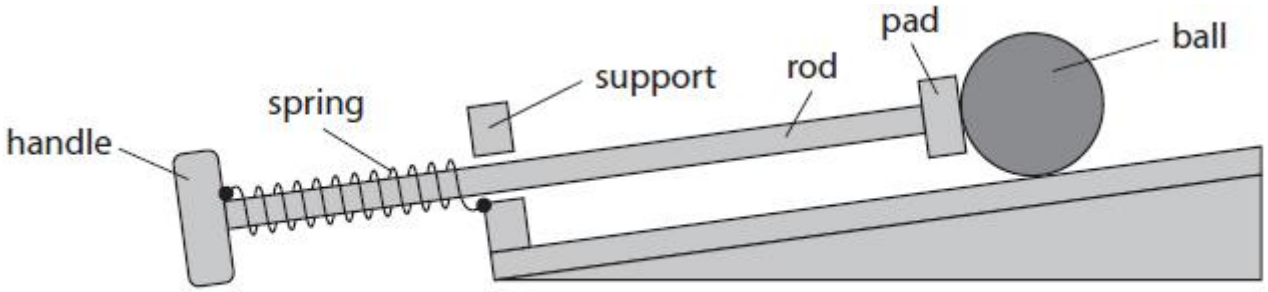


Figure 10

One end of the spring is fixed to the handle.

The other end of the spring is fixed to the support.

A child pulls the handle, stretching the spring.

Figure 11 shows the toy with the spring stretched.

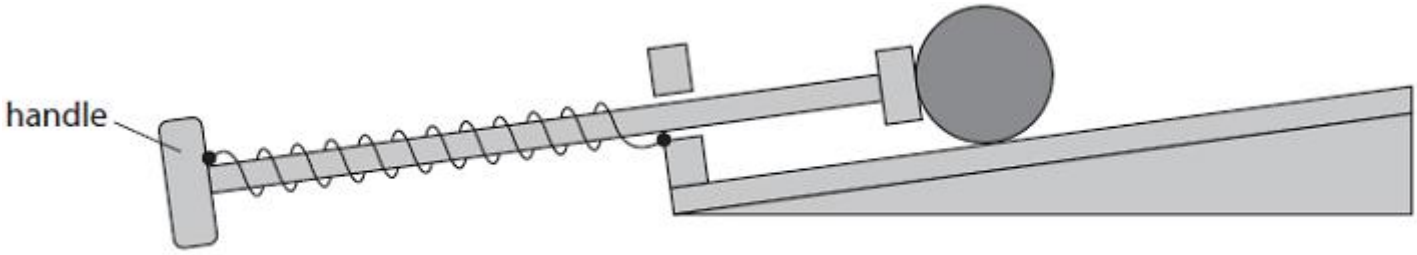


Figure 11

(i) Which of these shows the forces acting on the handle when the child keeps the spring stretched?

Ignore the force due to gravity.

(1)

☒ **A** ☒ **B** ☒ **C** ☒ **D**

(ii) In Figure 11, the extension of the spring is 0.070 m.

The spring constant (k) is 20 N/m.
Calculate the force used to extend the spring.
Use the equation

$$\text{force} = k \times \text{extension}$$

(2)

force = ... N

(Total for question = 3 marks)

Q11.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 1 shows an object under the surface of the sea.

(i) Which arrow shows where the pressure on the object is greatest?

(1)

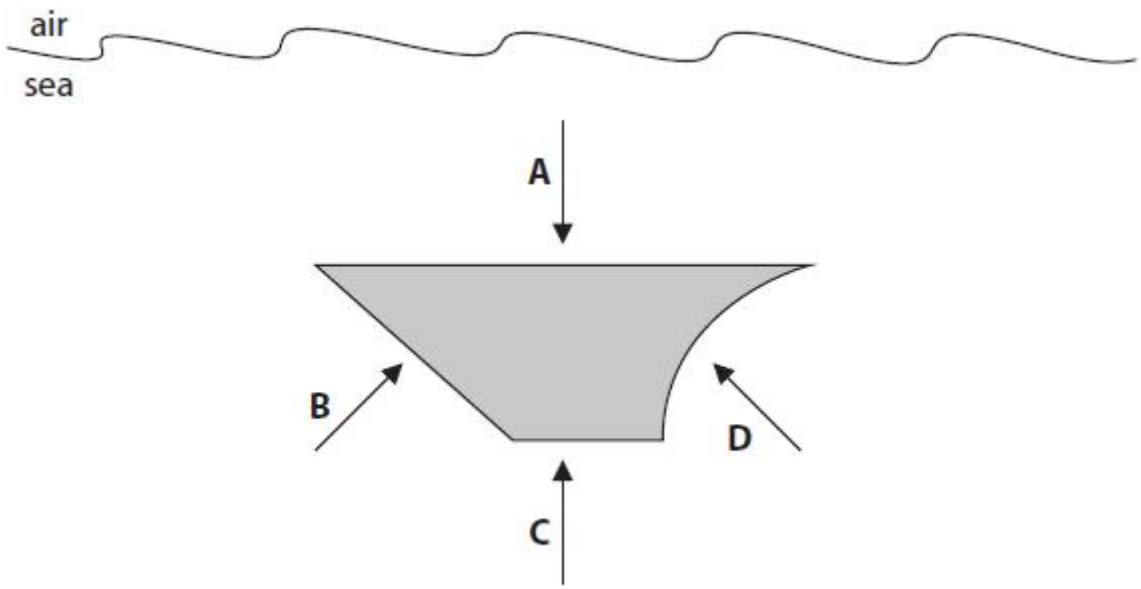


Figure 1

- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☐ **D**

(ii) Figure 2 shows two blocks of stone resting on the bottom of the sea.

Both blocks have the same height.
Area Y is 4 times bigger than area X.

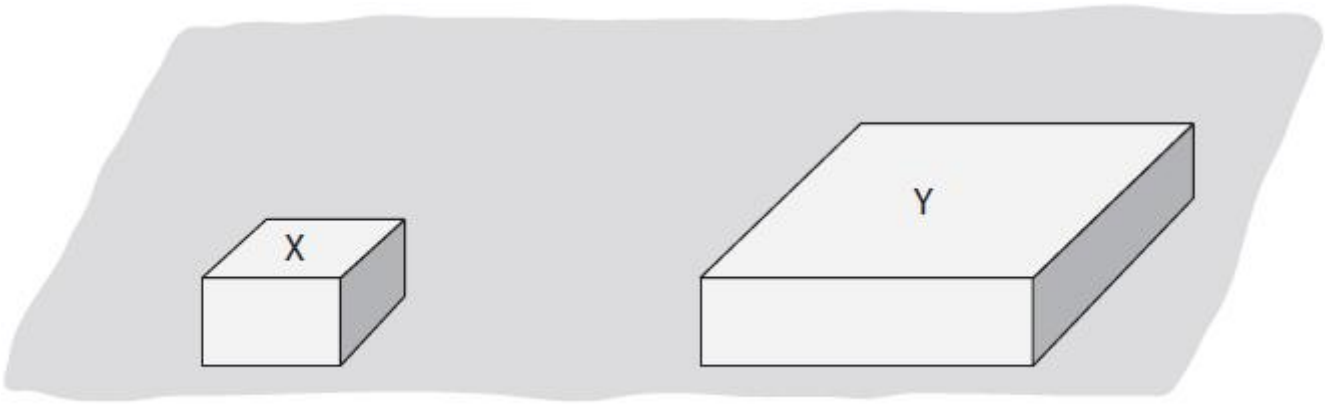


Figure 2

What is the pressure due to the water on the top surface of block Y?

(1)

- ☐ **A** one quarter of the pressure on X
- ☐ **B** the same as the pressure on X
- ☐ **C** twice the pressure on X
- ☐ **D** four times the pressure on X

(Total for question = 2 marks)

Mark Scheme

Q1.

Question	Answer	Mark
	 This shows the **only** direction normal to surface, acting towards surface	**(1)** **AO3.1**

Q2.

Question number	Answer	Additional guidance	Mark
	two similarities such as: (2) • both use the same loads • both start/end with same extension • both return to original length **two differences such as:** (2) • extensions for spring and rubber band differ • spring - loading and unloading are the same – rubber band different • extension- spring linear, rubber band non-linear	go up evenly/even steps steps uneven	**(4)**

Q3.

Question Number	Answer	Mark
	C 3 **C** is the only correct answer. **A** is incorrect because it does not include the pressure of the water above the diver. **B** is incorrect because it only includes the pressure of 10m of water above the diver. **D** is incorrect because it includes the pressure 0r 30m of water above the diver.	**(1)**

Q4.

Question number	Answer	Additional guidance	Mark
i	data points correctly identified (1) 50 ±2 80 ±2 evaluation (1) (-)30 ±4 kPa	award 1 mark if 80 and 50 seen ignore the lack of minus sign allow ecf from incorrect reading of either pressure at 2000m or pressure at 6000m for one mark	**(2)** **AO3**

Question number	Answer	Additional guidance	Mark
ii	any **one** suggestion of greater density of atmosphere (1) greater depth of atmosphere (above the aeroplane) (1) greater temperature (of the atmosphere) (1)	accept reverse argument more particles (per cubic metre) the air gets thicker greater weight of the atmosphere	**(1)** **AO1**

Q5.

Question Number	Answer	Additional guidance	Mark
	an explanation linking: (snowshoes have a) large(r) (surface) **area** (than boots) (1) reduces the pressure (on the snow) (1)	accept reverse argument throughout accept do not sink into snow/ stay on top of the snow	**(2)** **AO2.2**

Q6.

Question Number	Answer	Additional guidance	Mark
	An explanation to include the following MP1: (as the balloon rises) it gets bigger (1) Any **two** from: MP2: (because) density of air decreases / fewer (air) particles (in the atmosphere) (1) MP3: pressure (outside the balloon) decreases (1) MP4: pressure inside (balloon) is greater than pressure outside (1)	accept balloon bursts air gets thinner accept a named component of air Two from MP2, MP3 and MP4 can still be awarded even if MP1 is not.	**(3)**

Q7.

Question number	Answer	Mark
	An explanation that combines identification – application of knowledge (1 mark) and reasoning/justification – application of understanding (3 marks): • the gas cylinders have the same weight (1) • but cylinder A has the smallest area (that is in contact with the ground) (1) • the smaller the area, the greater the pressure (or reference to $P = \frac{F}{a}$) (1) • hence cylinder A exerts a greater pressure on the ground (1)	**(4)**

Q8.

Question	Answer	Additional guidance	Mark
		ignore gravity accept reverse arguments	**(2)** **AO3.2**
	an explanation linking:		
	density decreases as height increases (1)	density decreases as you go higher	
	with		
	(because) particles are further apart (higher up) (1)	accept fewer particles per unit volume accept particles more spaced out	

Q9.

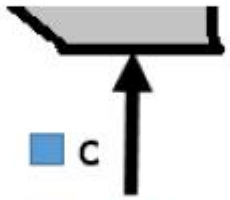
Question number	Answer	Additional guidance	Mark
	an explanation linking the area (of contact between person and bed) is smaller when standing up (1) same weight (over smaller area) so the pressure is greater when standing up (1)	accept reverse arguments weight is more concentrated / not distributed /not spread across bed (when standing up) uses p = F/A argument (as a consequence of the smaller area, pressure is bigger)	**(2)** **AO2**

Q10.

Question number	Answer	Additional guidance	Mark
i	 D A and B are incorrect because they only show one force C is incorrect because the forces are in the wrong direction		**(1)**

Question number	Answer	Additional guidance	Mark
ii	substitution (1) (F =) 20 x (0.0)7 evaluation (1) 1.4 (N)	award full marks for the correct answer without working allow 1 mark max for POT error	(2)

Q11.

Question number	Answer	Additional guidance	Mark
i	 A, B and D are incorrect because they are all closer to the surface		(1)

Question number	Answer	Additional guidance	Mark
ii	B. the same as the pressure on X A,C and D are incorrect because the pressure does not depend on surface area		(1)